

```
!pip -q install opencv-python-headless

import cv2
import numpy as np
import matplotlib.pyplot as plt
from google.colab import files

print("OpenCV version:", cv2.__version__)
```

OpenCV version: 5.0.0

```
uploaded = files.upload()

image_name = next(iter(uploaded.keys()))
image = cv2.imread(image_name)

if image is None:
    raise ValueError("The uploaded file is not a valid image.")

# Convert color image to grayscale
gray = cv2.cvtColor(image, cv2.COLOR_BGR2GRAY)

print("Image loaded successfully:", image_name)
print("Original image shape:", image.shape)
print("Grayscale image shape:", gray.shape)
```

tiger cub.jpg
tiger cub.jpg(image/jpeg) - 29175 bytes, last modified: 8/24/2026 - 100% done
Saving tiger cub.jpg to tiger cub.jpg
Image loaded successfully: tiger cub.jpg
Original image shape: (738, 416, 3)
Grayscale image shape: (738, 416)

```
plt.figure(figsize=(12, 5))

# Display grayscale image
plt.subplot(1, 2, 1)
plt.imshow(gray, cmap="gray")
plt.title("Original Grayscale Image")
plt.axis("off")

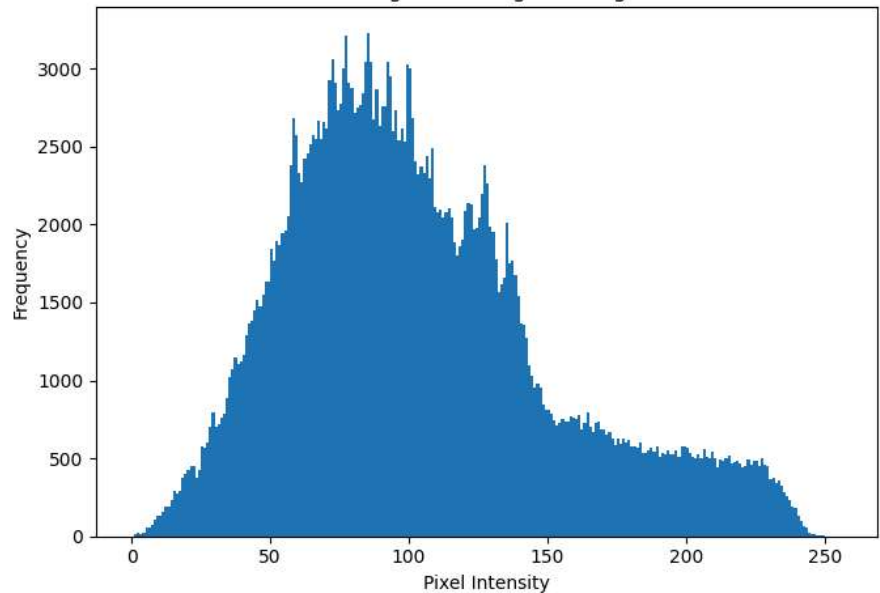
# Generate and display histogram
plt.subplot(1, 2, 2)
plt.hist(gray.ravel(), bins=256, range=[0, 256])
plt.title("Histogram of Original Image")
plt.xlabel("Pixel Intensity")
plt.ylabel("Frequency")

plt.tight_layout()
plt.show()
```

Original Grayscale Image



Histogram of Original Image



```
min_intensity = np.min(gray)
max_intensity = np.max(gray)

print("Minimum pixel intensity:", min_intensity)
print("Maximum pixel intensity:", max_intensity)
```

```
Minimum pixel intensity: 1
Maximum pixel intensity: 253
```

```
min_val= np.min(gray)
max_val=np.max(gray)

#Apply contrast stretching

contrast_stretched =cv2.normalize(
gray,None, 0, 255, cv2. NORM_MINMAX
)

print("Original minimum intensity:", min_val)
print("Original maximum intensity:", max_val)
print("After contrast stretching:")
print("Minimum intensity:", np.min(contrast_stretched))
print("Maximum intensity:", np.max(contrast_stretched))
```

```
#Display original and contrast-stretched images
```

```
plt.figure(figsize=(12, 5))
```

```
plt.subplot(1, 2, 1)
plt.imshow(gray, cmap="gray")
plt.title("Original Grayscale Image")
plt.axis("off")
```

```
plt.subplot(1, 2, 2)
plt.imshow(contrast_stretched, cmap="gray")
plt.title("Contrast Stretched Image")
plt.axis("off")
```

```
plt.tight_layout()
plt.show()
```

Original minimum intensity: 1
 Original maximum intensity: 253
 After contrast stretching:
 Minimum intensity: 0
 Maximum intensity: 255

Original Grayscale Image



Contrast Stretched Image



```
hist_equalized = cv2.equalizeHist(gray)
plt.figure(figsize=(12,5))

plt.subplot(1,2,1)
plt.imshow(gray, cmap="gray")
plt.title("Original Grayscale Image")
plt.axis("off")

plt.subplot(1,2,2)
plt.imshow(hist_equalized, cmap="gray")
plt.title("Histogram Equalized Image")
plt.axis("off")

plt.tight_layout()
plt.show()
```

Original Grayscale Image



Histogram Equalized Image



```
import cv2
import numpy as np
```

```
import numpy as np
import matplotlib.pyplot as plt

clahe = cv2.createCLAHE(clipLimit=2.0, tileGridSize=(8, 8))

clahe_img = clahe.apply(gray)

plt.figure(figsize=(5, 5))

plt.imshow(clahe_img, cmap="gray")
plt.title("CLAHE Enhanced Image")
plt.axis("off")

plt.show()
```

CLAHE Enhanced Image



```
import cv2
import numpy as np
import matplotlib.pyplot as plt

fig, axs = plt.subplots(1, 4, figsize=(18,5))

axs[0].imshow(gray, cmap='gray')
axs[0].set_title("Original")
axs[0].axis('off')

axs[1].imshow(contrast_stretched, cmap='gray')
axs[1].set_title("Contrast Stretched")
axs[1].axis('off')

axs[2].imshow(hist_equalized, cmap='gray')
axs[2].set_title("Histogram Equalized")
axs[2].axis('off')

axs[3].imshow(clahe_img, cmap='gray')
axs[3].set_title("CLAHE")
axs[3].axis('off')

plt.tight_layout()
plt.show()
```

Original



Contrast Stretched



Histogram Equalized



CLAHE

